Defect Detection in Remote Sensing Satellite Images: A New Dataset and Algorithm

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***Abstract*— Satellite observation is an important way to understand the earth. However, due to problems such as satellite aging, cloud obstruction, and other interferences during the imaging and transmission process, remote sensing images inevitably produce various defects. Hence, it is necessary to quickly detect defects to calibrate the imaging system and avoid the waste of satellite resources. Current research on defect detection in remote sensing images is not comprehensive and only focuses on partial defect categories, such as cloud and stripe. To this end, we construct the first large-scale high-resolution remote sensing image defect detection dataset (HRSD). The proposed dataset contains more than 1.2 million manually annotated patches from eight different satellites, covering various common defect categories and including multiple image modalities (i.e., panchromatic and multispectral). The dataset also has rich diversity which covers different landforms in multiple regions. Furthermore, to realize the detection of multiple defect categories simultaneously, we design a feature aggregation graph network (FAGN) based on the position correlation and semantic similarity among image patches, which fully utilizes the distribution characteristics of defects to achieve accurate defect detection. Extensive experiments on the HRSD dataset demonstrated the effectiveness of FAGN. We will release the HRSD dataset and FAGN model later.**

***Index Terms*— Dataset, graph network, remote sensing image defect detection, satellite observation.**

I. Introduction

With the rapid development of satellite observation technology and the increasing number of earth observation satellites, it is becoming an important way to understand the earth [1]. Earth observation satellites produce massive amounts of remote sensing images every day [2], [3]. These images have been widely used in many fields, such as precision agriculture [4], disaster management [5], urban planning [6], and environmental monitoring [7]. However, due to issues such as satellite aging, cloud obstruction, and other interferences during the imaging and transmission process, remote sensing images inevitably suffer from various defects, such as cloud, stripe defect, and pulse defects, as shown in Fig. 1(c). These defects will not only seriously affect the quality of remote sensing images, but also bring great difficulties to subsequent tasks such as image classification [8], object detection [9], or object recognition [10]. Therefore, it is necessary to quickly detect defects to correct the imaging system and avoid the waste of satellite resources.

In the remote sensing community, existing defect detection methods mainly focus on cloud/stripe detection. Most of the early methods [11], [12], [13], [14], [15] require manually designed features. For instance, Tsai and Chen [16] utilized the Canny algorithm to extract edge features from the image scene, and then employed the line-tracing algorithm to detect stripe noise. Başeski and Cenaras [17] used the color and texture features of the cloud regions to achieve cloud detection. However, the selection and design of manual features often rely on prior knowledge, making it difficult to effectively capture the complex patterns in remote sensing images, thereby limiting their performance. In recent years, deep learning methods have made significant progress in the field of computer vision, and these techniques have also been widely applied in the remote sensing community. Benefiting from their powerful feature learning capabilities, deep learning-based algorithms have achieved significant success in cloud/stripe detection [18], [19], [20], [21], [22], [23], [24]. For example, Yang et al. [25] proposed a cloud detection neural network that utilizes a feature pyramid module to extract multiscale features and a boundary refinement module to capture clear and detailed object boundaries for accurate cloud detection. Li et al. [23] proposed a linear stripe noise detection algorithm, which utilizes a deep convolutional neural network (CNN) based on YOLOv4 [26] to achieve superior stripe noise detection performance. However, the defect types in remote sensing images are diverse and their distributions are inhomogeneous. These methods only focus on specific defect categories (e.g., cloud, stripe) and ignore the spatial distribution correlation of defects, making it difficult to meet the requirements of practical applications.

Furthermore, large-scale and high-quality datasets are crucial for deep learning-based algorithms. Although some

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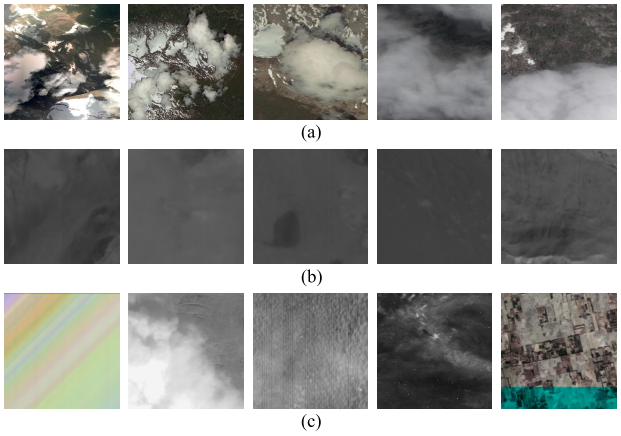


Fig. 1. Some examples taken from (a) cloud detection dataset, (b) stripe defect data of GF-1 satellite, and (c) proposed HRSD dataset. The defect types of the examples in (c) are stretch defect, cloud, stripe defect, pulse defect, and band anomaly. Most existing datasets only contain partial defects (cloud or stripe defect), such as (a) and (b), and do not cover most defect categories in remote sensing images. Therefore, we construct the HRSD dataset, covering various common defect types in remote sensing images, such as (c).

excellent defect datasets have been proposed in the remote sensing community, such as GF-1 WFV dataset [27], CSWV dataset [28], CloudS26 dataset [29], and SOSD dataset [24], these datasets only contain partial defects (cloud or stripe defect) and do not cover most defect categories in remote sensing images, as shown in Fig. 1. Moreover, most existing datasets have shortcomings such as small scale and insufficient image diversity. These limitations have seriously hindered the development of remote sensing image defect detection methods.

To address the above problems, we construct the first large-scale high-resolution remote sensing image defect detection dataset (HRSD). The proposed dataset contains more than 1.2 million manually annotated patches from eight different satellites, covering various common defect categories in remote sensing images. All patches are cropped from original remote sensing images. Each patch is 256 × 256 pixels in size and is annotated based on whether it contains a defect and what type of defect it contains. The annotation of each patch is completed by multiple individuals to mitigate personal biases. Moreover, the dataset has rich diversity which covers different landforms in multiple regions. The images in the HRSD dataset come from different types of sensors, including panchromatic and multispectral modalities.

Next, on the basis of the HRSD dataset, to realize the detection of defects quickly, i.e., defect localization and classification, we directly perform defect classification on image patches. Based on the above strategy, a feature aggregation graph network (FAGN) is proposed in this article for remote sensing image defect detection. First, FAGN constructs a spatial graph structure based on the positional correlation between patches within the neighborhood and simultaneously constructs a semantic graph structure based on semantic similarity. Then, a graph convolutional network (GCN) is utilized to aggregate spatial correlation features and semantic similarity features to achieve accurate defect detection. The encouraging experimental results presented in Section V demonstrate that our method is effective for remote sensing image defect detection.

In summary, the main contributions of this article are summarized as follows.

1) To our knowledge, the HRSD dataset is the first large-scale high-resolution remote sensing image defect detection dataset, covering various common defect categories and can be used as a benchmark dataset to promote the development of data-driven remote sensing image defect detection methods.
2) We propose a baseline model called FAGN, which fully utilizes the positional correlation and semantic similarity between patches to achieve accurate defect detection.

The rest of this article is organized as follows. In Section II, we introduce some related works about our work. Section III mainly introduces our proposed HRSD dataset. Section IV mainly introduces our proposed FAGN model. Section V displays the experimental results on HRSD datasets and analyzes the results. Finally, we draw conclusion in Section VI.

II. Related Work

A. Defect Detection Dataset in Remote Sensing Images

It is known that large-scale and high-quality datasets are crucial for deep learning-based algorithms. For remote sensing satellite images, clouds and stripes are common defects, and researchers have constructed some excellent cloud detection datasets and stripe detection datasets. For example, Li et al. [27] proposed the GF-1 WFV dataset, which contains 108 full scenes of level 2A with a spatial resolution of 16 m. As a typical remote sensing dataset for cloud detection, many methods have assessed their effectiveness on it. Subsequently, Zhang et al. [28] constructed the cloud and snow dataset CSWV based on WorldView2, which contains 27 WorldView2 images with clouds and snow. The spatial resolution of this dataset is 0.5∼10 m. Based on [28], Zhang et al. [29] proposed the CloudS26 dataset, which contains 26 RSIs with corresponding superpixel-level clouds and pixel-level annotations. The Levir_CS dataset proposed by Wu et al. [30] is also a cloud and snow coexistence dataset, which contains 4168 Gaofen-1 satellite images and corresponding geographical records. He et al. [31] created a cloud detection dataset AIR-CD, which contains 34 full scenes collected by the GF-2 satellite from different areas of China. The scenes in the AIR-CD dataset are complex and diverse, including a variety of land-cover types. Li et al. [23] constructed a large-scale remote sensing image stripe noise dataset through a linear transformation. The dataset contains 178 070 training images and 12 500 validation images.

In summary, the mentioned dataset only contains specific defects and does not cover most defect categories in remote sensing images. To this end, we construct a large-scale high-resolution remote sensing image defect detection dataset named HRSD. This dataset contains various common defect categories in remote sensing images, which is conducive to the research of remote sensing image defect detection algorithms.

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B. Defect Detection Method in Remote Sensing Images

In recent years, with the development of earth observation technology, the amount of remote sensing satellite data has increased explosively. Defects in remote sensing satellite images have received widespread attention from researchers, and some cloud detection algorithms and stripe detection algorithms have been successfully developed. For example, Shao et al. [32] proposed a model combining autoencode network and fuzzy function strategy for cloud detection in remote sensing images, achieving good performance. Subsequently, Yan et al. [33] proposed a multilevel feature fusion segmentation network, which fuses semantic information with different levels of spatial information to better handle multiscale objects. Shao et al. [34] proposed a model based on multiscale features, which combines high-level semantic information and low-level spatial information to greatly improve detection results. Luotamo et al. [35] proposed a network composed of two cascaded CNN models, in which the coarse component is used to classify patches as overcast, cloudless, or partly cloudy, and then the fine-grained component is used to classify pixels of "partly cloudy" patches. He et al. [31] proposed a lightweight network that improves the adaptive modeling capability of multiscale features and focuses on cloud boundary information, achieving high-precision detection of complex clouds. Zhang et al. [36] proposed a multiscale attention CNN to learn multiscale information through dilated convolution with different dilation rates, and use the attention module to highlight important features. Ma et al. [37] proposed a hybrid CNN-transformer network for cloud detection in remote sensing images. It effectively utilizes high-resolution local features and global information by combining the advantages of CNN and transformer. Li et al. [24] proposed a robust detection algorithm for oblique stripe noise, which adopted the backbone of the YOLOv5 [38] network model and used k-means clustering [39] to obtain multiple prior lengths suitable for stripe noise.

In a word, the mentioned algorithms are all aimed at a single defect, and are difficult to meet the detection needs of multiple defects in remote sensing images. To overcome this limitation, we propose a baseline model called FAGN to detect various defects existing in remote sensing images.

C. GCNs in Remote Sensing Community

GCN [40] is a deep learning method based on graph-structured data. Benefiting from their powerful representation capabilities, GCNs have been applied in many fields, such as recommender systems [41], traffic prediction [42], social analysis [43], and natural language processing [44].

In the remote sensing community, GCN has also become increasingly popular among researchers in recent years and has been widely used in various remote sensing tasks, such as remote sensing scene classification [45], [46] and hyperspectral image classification [47], [48], [49]. Mou et al. [48] proposed a graph-based semisupervised model named nonlocal GCN for hyperspectral image classification by utilizing both labeled and unlabeled data. He et al. [50] proposed a dual GCN for hyperspectral image classification with limited training samples, in which the first GCN is used to fully extract sample features, and the second GCN uses label distribution learning to reduce training sample requirements. Xu et al. [51] proposed a deep feature aggregation framework driven by GCN. It uses GCN to effectively exploit the patch-to-patch correlations of convolutional feature maps, providing more refined features for remote sensing scene classification. Gao et al. [46] proposed a high-order GCN for remote sensing scene classification, which learns the dependencies between scene semantic classes in different orders. Zhang et al. [52] proposed a model based on truncated sparse singular value decomposition and GCN to improve classification accuracy by leveraging the contextual relationship between adjacent objects. Yang et al. [53] proposed a semantic-aware graph network that uses a dynamic GCN-based graph node update strategy to capture pixel-level semantic and regional-level contextual information in remote sensing scenes. In this article, inspired by the above GCN, we explore a method of applying GCN to fully exploit the characteristics of defect distribution for remote sensing image defect detection.

III. Dataset

In recent years, with the development of satellite observation technology, the number of remote sensing images has grown explosively. Clouds and stripes are common defects in remote sensing images, and some excellent cloud detection datasets and stripe detection datasets have been successfully developed. However, these datasets only contain a part of the defect categories of remote sensing images, which seriously limits the development of data-driven remote sensing image defect detection. To address this problem, we propose the HRSD dataset, which is a large-scale high-resolution remote sensing image defect detection dataset that covers various common defect categories in remote sensing images.

A. Defect Type Selection

The first step in constructing a dataset is to select representative defect categories. To this end, we observe and analyze the defect categories existing in high-resolution remote sensing images, and summarize the defect categories into seven different defects: information loss, stretch defect, cloud, stripe defect, pulse defect, band anomaly, and other defects. The proposed HRSD dataset contains the above seven defect categories and no defects category. The example of each category in the HRSD dataset is shown in Fig. 2.

1) Information Loss: Information loss is mainly caused by satellite aging or problems in the transmission process. In remote sensing images, information loss usually means that there are large areas without texture details and appear as pure black.

2) Stretch Defect: Due to the influence of terrain, especially the drastic changes in terrain, it is impossible to ensure that every ground target point can be imaged on the satellite sensor, resulting in texture stretching in some areas with insufficient information, causing image texture distortion.

3) Cloud: Cloud is a common climate phenomenon, covering more than 60% of the earth's area. In remote sensing images, clouds have a variety of forms, and different forms are determined by different formation causes and processes.

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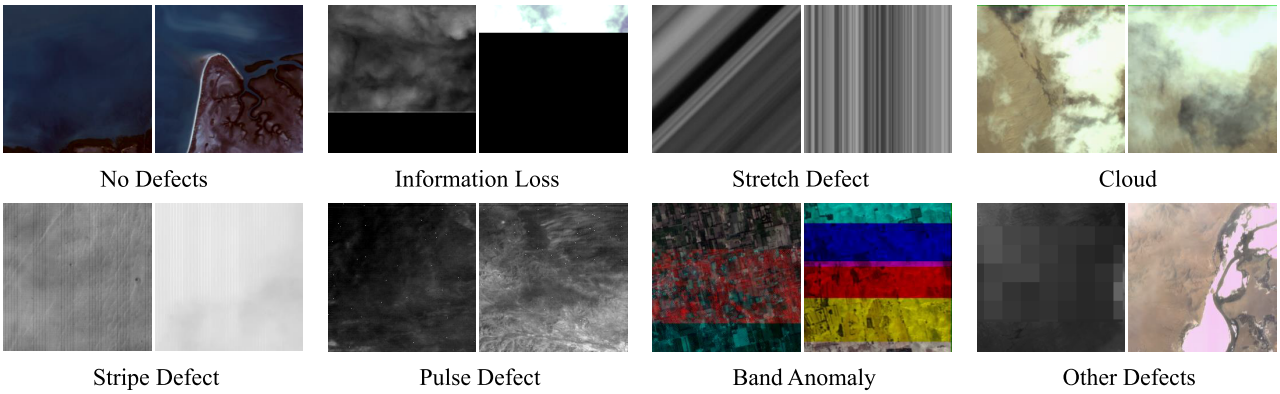


Fig. 2. Some examples of the proposed HRSD dataset. This dataset is acquired from eight different satellites and covers almost all defect categories in remote sensing images.

TABLE I
DETAILS OF THE ORIGINAL REMOTE SENSING SATELLITE IMAGES IN THE HRSD DATASET

Satellite	GF1	GF2	GF4	GF5	SJ9A	SV1	ZY1	ZY3
Panchromatic Spatial Resolution (m)	2	1	50	50	2.5	0.5	2.3	2.1
Panchromatic Image Number	59	16	8	12	9	12	29	8
Multispectral Spatial Resolution (m)	8	4	50	/	10	2	/	5.8
Multispectral Image Number	74	19	8	0	9	12	0	29
Total Images	133	35	16	12	18	24	29	37
Image Size	$4k \times 4k$ ~ $35k \times 18k$	$7k \times 7k$ ~ $24k \times 26k$	$1k \times 1k$ ~ $10k \times 10k$	$1k \times 1k$	$3k \times 3k$ ~ $12k \times 12k$	$0.5k \times 0.5k$ ~ $25k \times 25k$	$24k \times 24k$ ~ $25k \times 12k$	$8k \times 8k$ ~ $25k \times 25k$

4) Stripe Defect: Stripe defect is mainly caused by satellite aging and other interference. In the original image data, stripe defects are usually distributed horizontally or vertically and have a certain spatial width. Moreover, the brightness of stripe defects varies and can be divided into bright stripes and dark stripes.

5) Pulse Defect: Pulse defect is generally caused by damage to some pixels of satellite sensors, which will produce random black and white noise in remote sensing images and greatly affect the image quality.

6) Band Anomaly: Band anomaly is caused by data loss that occurs during the transmission of original data captured by satellites, which will produce bands of multiple colors in the remote sensing images.

7) Other Defects: In addition to the above six common defect types, there are some other defect types that are characterized by small quantity, messy, and irregular. These defect types are collectively counted as other defects.

The mentioned seven defect categories cover various common defect categories in remote sensing images, and can provide a rich variety of data for remote sensing image defect detection research.

B. Dataset Collection

The proposed HRSD dataset is sourced from 304 original high-resolution remote sensing defect images acquired from eight different satellites. The eight satellites are GF1, GF2, GF4, GF5, SJ9A, SV1, ZY1, and ZY3. The HRSD dataset contains a variety of different sensor data, greatly expanding the richness of the data. The images in this dataset have two modalities: panchromatic and multispectral. The number of images provided by each satellite varies from 12 scenes to 133 scenes, and the image shooting time ranges from 2013 to 2019. The HRSD dataset is multiresolution, with pixel resolutions ranging from 0.5 to 50 m. Details of the original high-resolution remote sensing defect images are shown in Table I. Besides, there may be multiple defects to coexist in an original image. Table II statistics the data volume corresponding to different numbers of defects. These original images are used as source data to construct the HRSD dataset.

TABLE II
DATA VOLUME CORRESPONDING TO DIFFERENT NUMBERS OF DEFECTS

Defect Number	0	1	2	3	4	5	6	7
Image Number	12	83	87	80	39	3	0	0

The original high-resolution remote sensing images of the HRSD dataset are large, some even reaching 25 000 × 25 000 pixels. It is difficult to directly calculate and process such large-size original high-resolution images. Moreover, when directly annotating defects on the original image, the workload

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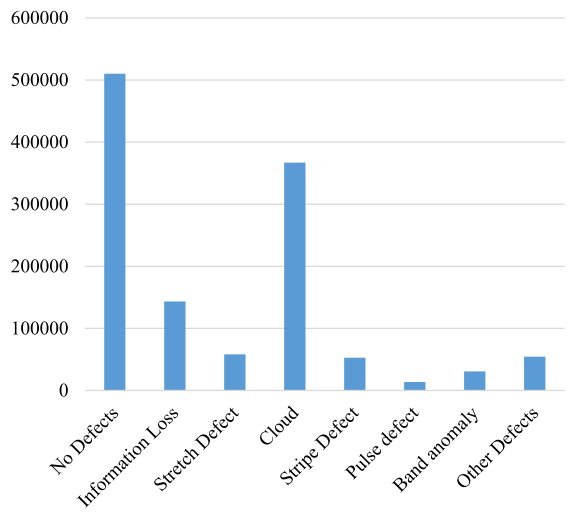


Fig. 3. Data volume corresponding to each category in the HRSD dataset.

of pixel-level annotation is too large and is impractical. Using bounding boxes for annotation is also difficult to calculate because the defect area in some images is too large. To this end, when constructing the dataset, we cropped the large-sized original remote sensing images into several small-sized patches, which is convenient for computational processing and can also make more efficient use of image information.

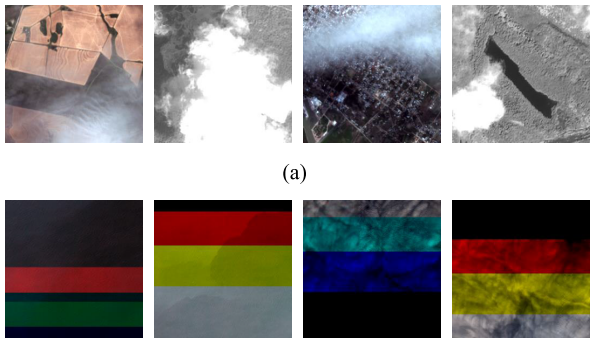
We crop all original remote sensing images in order from left to right and top to bottom, and record the row and column positions of the cropped patches in the original remote sensing images. The crop size is 256 × 256 pixels with a 20-pixel overlap between adjacent patches. When cropping to the rightmost column or bottommost row is less than 256 pixels, cropping is from right to left or from bottom to top.

After cropping the original remote sensing images, we need to annotate each patch. First, we divided the HRSD dataset into several non-overlapping subsets according to the original images, and each person was responsible for annotating the patches of a subset. After the patch annotation was completed, it was handed over to others for inspection and review. If there was a controversial annotation, multiple people voted, and the category with the most votes was the final category.

C. Dataset Characteristics

The proposed HRSD dataset is the first large-scale and publicly available high-resolution remote sensing image defect detection dataset. Fig. 3 reports the number of image patches for each category in this dataset. By observing Fig. 3, we can see that the number of patches of "no defects" and "cloud" is relatively large. Compared with other datasets, our proposed HRSD has the following three salient characteristics.

1) Large Scale: The HRSD dataset contains 1 203 587 high-resolution remote sensing defect patches and 1 230 112 manually annotated labels, of which 26 525 patches contain two defect categories. These patches are divided into eight categories, including no defects and seven defect categories.



(a)

(b)

Fig. 4. High intra-class diversity. Different patches in the same category may have significant differences. (a) Cloud. (b) Band anomaly.

The HRSD dataset covers various common defect categories in remote sensing images. To our knowledge, the HRSD dataset is the first large-scale high-resolution remote sensing image defect detection dataset.

2) Intra-Class Diversity: The patches in the HRSD dataset have rich intra-class diversity, which makes it challenging. As shown in Fig. 4(a), the cloud layers contained in different patches in the cloud category are very different in shape and thickness, and the distribution of cloud layers in the patches is basically irregular. From Fig. 4(b), we can see that different patches in band anomaly contain bands that may have different colors and may differ significantly from each other. In addition, patches of the same defect category may be taken from different satellites, further enriching the degree of intraclass diversity of patches in the HRSD dataset.

3) Background Richness: The background of the patches in the HRSD dataset covers various scenes. The background comes from different landforms captured in multiple regions. It has four first-level categories and 12 second-level categories, i.e., farmland (flat farmland, terrace), city (strip-like city, point-like city, block-like city), waters (ocean, river, lake), and natural areas (Gobi, plain, desert, mountain). Examples of each background are shown in Fig. 5.

D. Comparison With Other Datasets

In this section, some related remote sensing image defect detection datasets (e.g., GF-1 WFV [27], CSWV [28], Levir_CS [30], AIR-CD [31], and SOSD [24]) are compared with the proposed HRSD dataset, as shown in Table III. These related datasets are briefly reviewed below.

1) GF-1 WFV: The GF-1 WFV dataset consists of 108 2A-level full-scene images and their corresponding pixel-level annotations, which are widely used in remote sensing image cloud detection tasks. The images in the dataset have a spatial resolution of 16 m and contain three visible bands and one near-infrared band.

2) CSWV: The CSWV dataset is sourced from the World-View2 satellite and contains 27 images with clouds and snow. The spatial resolution of the dataset is 0.5∼10 m. The diversity of clouds and snow in the dataset makes it more representative and generalizable.

3) Levir_CS: The Levir_CS dataset collects 4168 GF-1 WFV scenes, covering different ground feature types and

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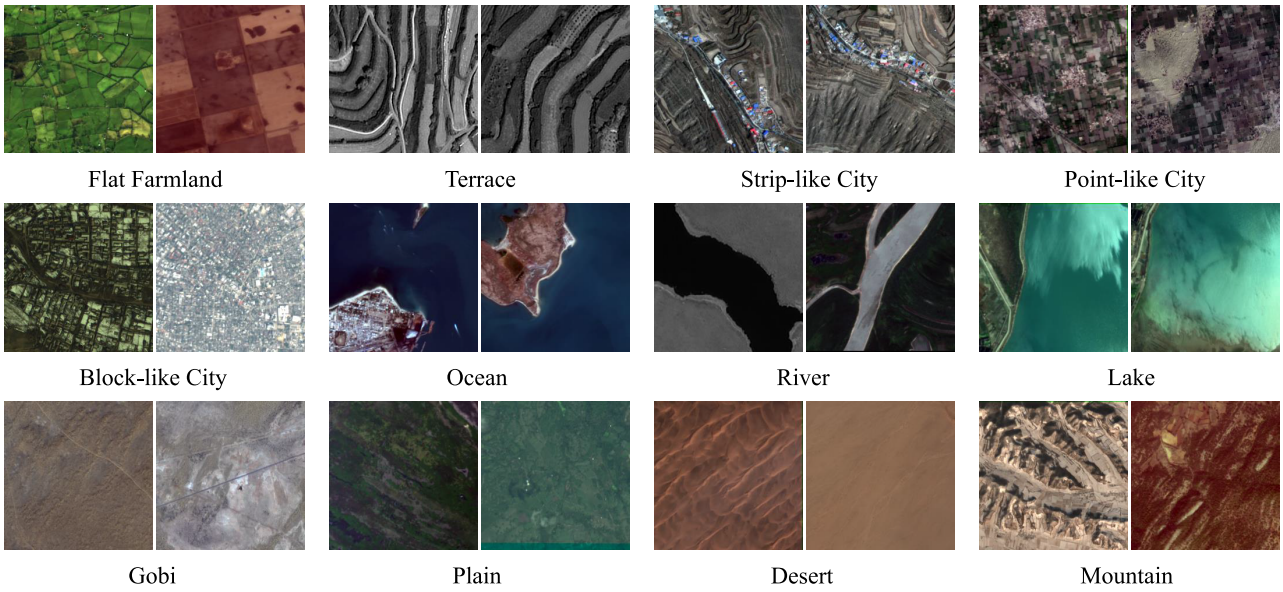


Fig. 5. Some examples of backgrounds in the HRSD dataset. The backgrounds of the images in this dataset come from different landscapes captured in multiple regions.

TABLE III
COMPARISON BETWEEN THE PROPOSED HRSD DATASET AND SOME EXISTING REMOTE SENSING IMAGE DEFECT DETECTION DATASETS

Dataset	Total images	Image size	Spatial resolution (m)	Defect type	Annotation type	Source
GF-1 WFV [27]	108	/	16	Cloud	pixel-level	GF-1
CSWV [28]	27	/	0.5∼10	Cloud,Snow	pixel-level	WorldView2
Levir_CS [30]	4168	1320 × 1200	160	Cloud,Snow	pixel-level	GF-1
AIR-CD [31]	34	7300 × 6908	4	Cloud	pixel-level	GF-2
SOSD [24]	157547	640 × 640	/	Stripe defect	object-level	GF-1,GF-2
HRSD(ours)	1203587	256 × 256	0.5∼50	Information loss,Stretch defect, Cloud,Stripe defect,Pulse defect, Band anomaly,Other defects	image-level	GF-1, GF-2, GF-4, GF-5,SJ-9A,SV-1, ZY-1,ZY-3

climate conditions. The image size in the dataset is 1320 × 1200, the spatial resolution is 160 m, and the annotation type is pixel-level annotation. In addition, the dataset also contains geographic information such as longitude, latitude, and altitude.

4) AIR-CD: The AIR-CD dataset is sourced from the GF-2 satellite and consists of 34 images and their corresponding pixel-level annotations. The image size in the dataset is 7300 × 6908 pixels, with a spatial resolution of 4 m, and includes visible and near-infrared bands. The dataset contains a variety of land cover types and the scenes are complex and diverse, making it more challenging.

5) SOSD: The SOSD dataset sources from the GF-1 and GF-2 satellites and contains 157 547 simulated stripe noise images with a size of 640 × 640 pixels. The labels of the dataset are the center point coordinates, length, and angle of the stripe noise in the image.

In summary, the advantages of our proposed HRSD dataset can be summarized as follows.

1) Multiple Defect Types: The proposed HRSD dataset comprises seven types of defects, which is more extensive than other datasets, thus supporting a broader range of application scenarios.

2) Rich Data Sources: The proposed HRSD dataset is sourced from eight different satellites, which is richer than the sources of other datasets and increases the diversity of data.

3) Large Data Scale: The proposed HRSD dataset contains a greater number of images compared to other datasets, better meeting the requirements of deep learning-based remote sensing image defect detection algorithms.

IV. METHOD

In this section, we introduce the proposed FAGN model in detail. The overall structure is shown in Fig. 6. Given $M \times M$ adjacent patches, we utilize a CNN to extract features for each patch. Then, we establish position nodes based on the position information of the patches and semantic nodes based on the predicted class information of patches. Next, a spatial graph structure is constructed based on the position correlation between patches in the neighborhood, and a semantic graph structure is constructed based on the semantic similarity. Finally, GCN is used to aggregate the spatial correlation features and semantic similarity features for accurate defect detection.

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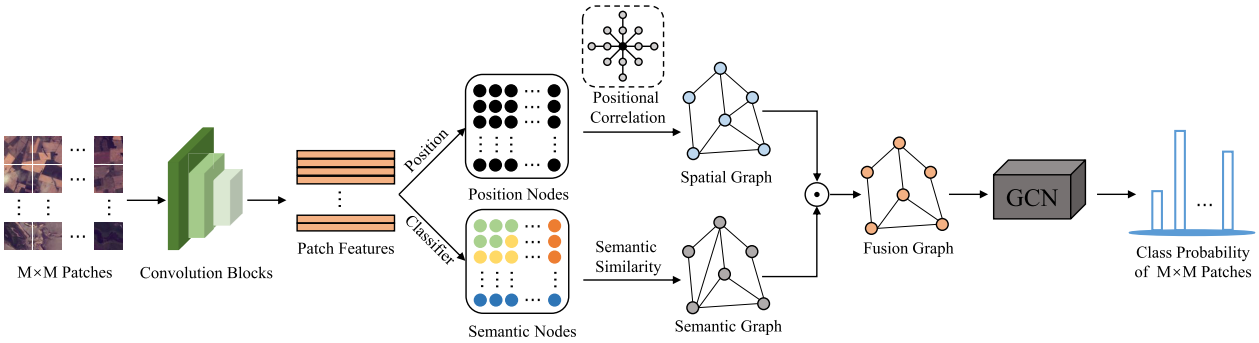


Fig. 6. Overall structure of our proposed FAGN. First, CNN is used to extract features for each patch. Then, the position nodes are established based on the position information of patches, and the semantic nodes are generated based on the defect class of each patch. The defect class is predicted by an additional MLP-based classifier. Next, a spatial graph structure is constructed by considering the position correlation, and a semantic graph structure is constructed by considering the semantic similarity. Finally, GCN is used to aggregate spatial correlation features and semantic similarity features to achieve accurate defect detection.

A. Positional Correlation-Based Graph Construction

Since the number of patches obtained by cropping an original image is huge, and not all patches are relevant. Therefore, it is more reasonable to establish a local graph structure than a global graph structure. In addition, establishing a local graph structure can reduce the size of the graph, which is beneficial to reducing the amount of calculation and improving the speed of model detection.

Through the observation of original remote sensing images, we found that defects are usually distributed horizontally, vertically, and regionally. When a defect appears in a patch, there is a high probability that the same defect will appear in the neighborhood. Therefore, we regard each patch as a node and select nodes in the horizontal, vertical, and diagonal directions of each node to construct the edges in the graph structure, as shown in the position correlation in Fig. 6.

According to the above method, we construct a local, fixed undirected graph structure for a local area composed of $M \times M$ patches, where the features extracted by CNN are used to represent each patch. The undirected graph structure can be represented by $G = (V, E)$, where V represents the set of graph nodes and E represents the edge set. The mathematical formulation of V and E are as follows:

$$V = (v_1, v_2, v_3, \ldots, v_N) \tag{1}$$

$$E = \{(v_1, v_2), (v_1, v_3), \ldots, (v_{N-1}, v_N))\} \tag{2}$$

where $N = M \times M$. Then we construct an adjacency matrix $A_p \in R^{N \times N}$ based on the undirected graph structure G. The calculation details of the adjacency matrix A_p can be summarized by the following formulation:

$$A_p = \{a_{ij}\}, a_{ij} = \begin{cases} 1, & \text{if } (v_i, v_j) \in E \\ 0, & \text{otherwise.} \end{cases} \tag{3}$$

It is worth noting that when constructing the undirected graph structure G, we choose to select $M \times M$ adjacent patches from left to right and from top to bottom in the original image. When the number of remaining rows or columns is less than M, select from bottom to top or from right to left.

B. Semantic Similarity-Based Graph Construction

The graph structure constructed based on positional correlation may have the problem that the categories of the nodes in the neighborhood are inconsistent with the central node. This problem will seriously harm the performance of the learned features. To address this problem, we propose a semantic similarity-based graph construction method to improve the positional correlation-based graph structure by introducing the semantic information of patches.

Since the trained CNN has learned rich semantic information about patches, we utilize CNN to generate a semantic label for each patch. The calculation details can be formulated as follows:

$$\hat{y}_i = f(F_i; W^f) \tag{4}$$

$$p_i = \text{argmax}(\hat{y}_i) \tag{5}$$

where f represents the classifier, which consists of a fully connected layer, F_i represents the feature extracted by CNN for the ith patch, W^f represents the parameters of the classifier, $\hat{y}_i$ denotes the class probability generated by the classifier f for the ith patch, and p_i means the semantic label of the ith patch.

Based on the semantic labels of nodes in the graph structure G, we can obtain a binary matrix $A_s \in R^{N \times N}$. If the semantic labels of two nodes are the same, the corresponding element in the binary matrix A_s is 1, otherwise it is 0. The calculation process of the binary matrix A_s can be summarized as the following formulation:

$$A_s = \{b_{ij}\}, b_{ij} = \begin{cases} 1, & \text{if } p_i = p_j \\ 0, & \text{otherwise.} \end{cases} \tag{6}$$

Through the semantic similarity in the binary matrix A_s, we can alleviate the problem of category inconsistency in the adjacency matrix A_p constructed based on positional correlation. Finally, the adjacency matrix A can be obtained by the following formulation:

$$A = A_p \odot A_s \tag{7}$$

where $\odot$ denotes the element-wise product.

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C. Feature Association

Graph convolution operation can help explore the deep relationship between different patches. To this end, we introduce GCN [40] to effectively aggregate the spatial correlation features and semantic similarity features of patches in the neighborhood. The input to the GCN consists of the features extracted by CNN for $M \times M$ patches and the adjacency matrix A. To retain the characteristics of the node itself, we add the identity matrix I_N to the graph adjacency matrix A to obtain a new adjacency matrix $\widetilde{A}$. The mathematical formulation of the adjacency matrix $\widetilde{A}$ is as follows:

$$\widetilde{A} = A + I_N. \tag{8}$$

Then, we compute the degree matrix $\widetilde{D}$ of the graph based on $\widetilde{A}$. The calculation details of the degree matrix can be formulated as follows:

$$\widetilde{D_{ii}} = \sum_{j=1}^{N} \widetilde{A_{ij}}. \tag{9}$$

By introducing the learnable weight W, the graph convolution state update process can be formulated as follows:

$$H^{(l)} = \sigma\left(\hat{A} H^{(l-1)} W^{(l)}\right) \tag{10}$$

where $H^{(l)}$ and $H^{(l-1)}$ are the output and input of the lth layer, $W^{(l)}$ denotes the trainable parameters of the lth layer, σ denotes the activation function. The calculation details of $\hat{A}$ can be formulated as follows:

$$\hat{A} = \widetilde{D}^{-1/2} \widetilde{A} \widetilde{D}^{-1/2}. \tag{11}$$

D. Loss Function

The activation function of the last layer of GCN is the Sigmoid function, which generates the predicted probability of each patch. We select the binary cross-entropy function as the loss function, and its calculation process is as follows:

$$L = -\frac{1}{N} \sum_{i=1}^{N} \frac{1}{C} \left(y_i \log\left(\hat{y_i}\right) + (1 - y_i) \log\left(1 - \hat{y_i}\right)\right) \tag{12}$$

where N is the number of patches, C is the number of patch categories, $\hat{y_i}$ is the predicted probability of the ith patch, y_i is the true label of the ith patch and the form is one-hot encoding.

V. Experiments

A. Experimental Settings

In this article, all experiments were conducted by Pytorch [54] in the computing environment of NVIDIA GeForce RTX 3090 and Intel[1] Core[2] i9-10900K CPU. In detail, the CNN in the FAGN model uses DenseNet-201 [55] pretrained on ImageNet [56], and the graph convolution process in the FAGN model is completed by a three-layer GCN. The input data of the three-layer GCN in each iteration is the feature vector of 12×12 patches.

In the training stage, we use a two-stage training strategy to train our FAGN model. In the first stage, DenseNet-201 is trained for a total of ten epochs, with the learning rate and batchsize set to 0.001 and 32, respectively. In the second stage, a three-layer GCN is trained for a total of 100 epochs, with the learning rate set to 0.0001. Here, we use the Adam optimization algorithm [57] to complete the training process of the FAGN model.

The data augmentation (including horizontal flipping and vertical flipping) is employed to enhance the performance of the model. In addition, for the HRSD dataset, we select patches from 10% or 20% of the original remote sensing images to construct the training set, and the remaining patches are used as the test set.

To verify the effectiveness of our FAGN model, we compare it with several excellent comparison algorithms, including three CNN models (GoogLeNet [58], VGGNet-16 [59], and ResNet50 [60]); two transformer models (Vit-base [61], PVT [62]); two GCN-based models (SAGN [53], PFFGCN [63]); and six CNN-based models (MF^{2}CNet [64], GRN-SNDL [65], SE-Net [66], SKAL [67], LCNN-BFF [68], and FSCNet [69]).

B. Evaluation Metrics

To fairly compare the performance of the FAGN model and the compared methods, we adopt overall accuracy (OA) and average accuracy (AA) as evaluation metrics. Here, OA is defined as the number of correctly predicted labels divided by the number of labels contained in the total test image, and AA is defined as the AA of all categories.

Suppose there is a test set $D = \{(x_i, y_i) | 1 \le i \le N\}$, where the binary vector $y_i = [y_{i1}, y_{i2}, \ldots, y_{iC}] \in [0, 1]^C$ is the ground-truth label of the ith test patch. It is worth noting that if the ith patch belongs to two categories, the corresponding prediction vector $\hat{y_i}$ sets the two categories with the largest values to 1 in the one-hot process. If the ith patch only belongs to one category, the corresponding prediction vector $\hat{y_i}$ sets the category with the largest value to 1 in the one-hot process. Then the OA and AA are defined as follows:

$$\mathrm{OA} = \frac{\sum_{i=1}^{C} {N_i}^t}{\sum_{i=1}^{C} {N_i}^l} \tag{13}$$

$$\mathrm{AA} = \frac{1}{C} \sum_{i=1}^{C} \frac{{N_i}^t}{{N_i}^l} \tag{14}$$

where ${N_i}^t$ denotes the number of correctly predicted labels for the ith category, ${N_i}^l$ denotes the number of labels contained in the ith category test image, and C denotes the total number of categories in the HRSD dataset. It is worth noting that the two evaluation metrics used in the experiments range from 0 to 1, and higher values indicate better performance.

C. Experimental Results

In this section, we report the results of our FAGN model and the compared methods on the HRSD dataset. The OA values of the FAGN model and the compared methods on the HRSD dataset are summarized in Table IV. In this table, the best results are shown in bold. OA(1:9) in this table means that the ratio between the training set and the test set is 1:9, and the

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TABLE IV
OA OF DIFFERENT METHODS ON HRSD DATASET

Method	OA (1:9)	OA (2:8)
GoogLeNet [58]	84.85	86.95
VGGNet-16 [59]	84.67	87.19
ResNet50 [60]	83.40	84.93
Vit-base [61]	85.85	87.33
PVT-S [62]	85.69	88.03
MF^{2}CNet [64]	70.37	78.83
GRN-SNDL [65]	86.16	88.72
SE-Net [66]	73.58	79.47
SKAL [67]	80.93	83.56
LCNN-BFF [68]	83.19	85.50
SAGN [53]	82.81	87.21
PFFGCN [63]	84.32	87.22
FSCNet [69]	85.41	87.98
FAGN(Ours)	**86.54**	**90.13**

same is true for OA(2:8). By observing Table IV, we can find that the performance of the FAGN model outperforms all compared methods on the two training sets of the HRSD dataset. When the ratio of the training set to the test set is 1:9, FAGN achieves 86.54%, outperforming the leading CNN model GoogLeNet (84.85%) and the top transformer model Vit-base (85.85%) by 1.69% and 0.69%, respectively. Compared to the best GCN-based model PPFGCN (84.32%), FAGN improved by 2.22%. FAGN outperformed the best CNN-based model GRN-SNDL (86.16%) by 0.38%. When the ratio of training set to test set is 2:8, FAGN achieves an encouraging 90.13% OA, which exceeds the best CNN-based model GRN-SNDL (88.72%) by 1.41% and the best GCN-based model PFFGCN (87.22%) by 2.91%. Compared with the top-scoring CNN model VGGNet-16 (87.19%), FAGN achieved an improvement of 2.94%. FAGN outperformed the best transformer model PVT-S (88.03%) by 2.10%.

In addition to the OA value, we also provide the accuracy of each category in the HRSD dataset and their average result AA. Table V summarizes the per-class accuracy and AA of the FAGN model and compared methods when the ratio of the number of train data to test data is 1:9. As in Table IV, the best results in Tables V are shown in bold. According to Table IV, it is easy to find that FAGN achieves the highest AA, reaching 67.66%. This result surpasses the best CNN model VGGNet-16 (66.71%) and the leading transformer model PVT-S (66.49%) by 0.95% and 1.17%, respectively. It also outperforms the best-performing CNN-based model FSCNet (67.08%) by 0.58%. Compared with the highest scoring GCN-based model SAGN (67.52%), FAGN improves by 0.14%. Besides, FAGN achieves excellent performance in many categories. For example, FAGN achieves 91.50% accuracy on "no defects," which exceeded the closest CNN model GoogLeNet (90.32%) by 1.18%. It also surpassed the highest-scoring transformer model Vit-base (89.94%) by 1.56%. Compared with the best GCN-based model PFFGCN (88.62%), FAGN improved by 2.88%.

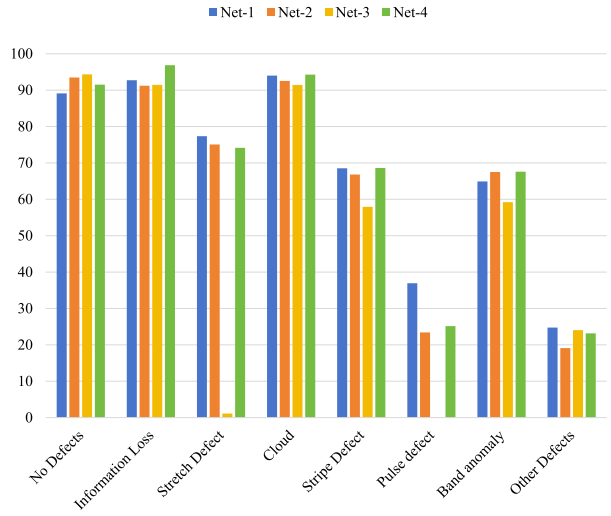


Fig. 7. Ablation experiment results for the HRSD dataset with a training ratio of 10%. Net-1∼Net-4 are the same as mentioned in Section V-D.

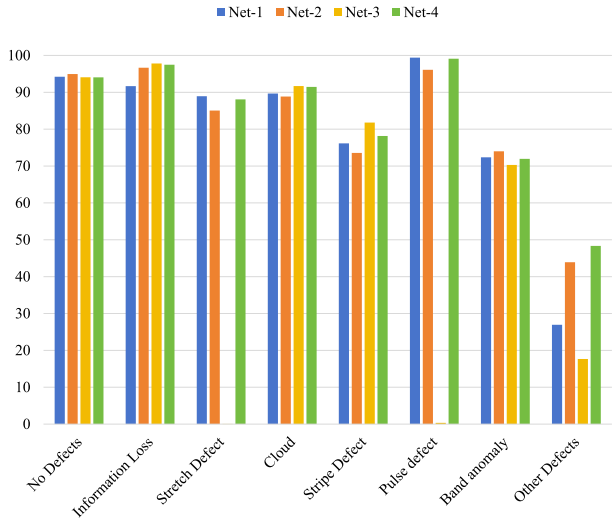


Fig. 8. Ablation experiment results for the HRSD dataset with a training ratio of 20%. Net-1∼Net-4 are the same as mentioned in Section V-D.

Table VI summarizes the per-class accuracy and AA of the FAGN model and compared methods when the ratio of the number of train data to test data is 2:8. As in Tables IV and V, the best results in Table VI are also shown in bold. From Table VI, it can be observed that FAGN achieved the highest AA, reaching 83.58%, which is superior to the best-performing CNN-based model FSCNet (79.62%) and the top-scoring GCN-based model SAGN (79.67%) by 3.96% and 3.91%, respectively. Compared to the top CNN model VGGNet-16 (82.10%) and the leading transformer model (79.48%), FAGN improves by 1.48% and 4.10%, respectively. Moreover, Table VI also shows that FAGN performs well in most categories, especially in "no defects," achieving an impressive accuracy of 94.06%, significantly surpassing the best performing GCN-based model PFFGCN (91.18%) by 2.88%. Compared with the highest-scoring CNN-based model GRN-SNDL (92.15%), FAGN improves by 1.91%. FAGN also surpasses the best CNN model GoogLeNet (91.70%) and the leading transformer model PVT-S (92.41%) by 2.36% and 1.65%, respectively.

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TABLE V
ACCURACY OF EACH CATEGORY OF DIFFERENT METHODS ON THE HRSD DATASET WHEN THE TRAINING RATIO IS 10%

Method	No Defects	Information Loss	Stretch Defect	Cloud	Stripe Defect	Pulse Defect	Band Anomaly	Other Defects	AA
GoogLeNet [58]	90.32	92.38	81.54	93.39	59.34	14.71	53.91	21.08	63.33
VGGNet-16 [59]	85.99	97.61	81.56	94.00	70.66	14.94	66.91	21.97	66.71
ResNet50 [60]	86.01	92.41	76.26	93.35	64.76	**34.05**	56.99	22.86	65.84
Vit-base [61]	89.94	97.32	83.76	94.13	64.10	13.65	61.36	18.90	65.39
PVT-S [62]	88.84	97.70	**83.82**	94.05	64.43	15.33	66.53	21.24	66.49
MF^{2}CNet [64]	73.86	92.19	56.59	81.86	1.00	10.99	49.79	12.71	47.37
GRN-SNDL [65]	88.09	**98.10**	78.93	92.70	51.35	32.92	66.24	23.72	66.51
SE-Net [66]	75.29	87.76	60.63	91.67	14.73	12.73	42.04	4.35	48.65
SKAL [67]	82.40	95.21	75.04	93.18	45.08	16.39	56.28	20.19	60.47
LCNN-BFF [68]	86.84	92.18	72.46	94.27	59.15	12.95	57.20	19.48	61.82
SAGN [53]	82.01	96.24	76.50	93.35	**74.54**	26.10	**67.82**	23.63	67.52
PFFGCN [63]	88.62	92.75	80.81	93.30	55.93	17.02	60.18	**24.75**	64.17
FSCNet [69]	87.16	97.27	81.59	**95.11**	67.11	17.49	67.37	23.52	67.08
FAGN(Ours)	**91.50**	96.86	74.15	94.27	68.60	25.14	67.58	23.15	**67.66**

TABLE VI
ACCURACY OF EACH CATEGORY OF DIFFERENT METHODS ON THE HRSD DATASET WHEN THE TRAINING RATIO IS 20%

Method	No Defects	Information Loss	Stretch Defect	Cloud	Stripe Defect	Pulse Defect	Band Anomaly	Other Defects	AA
GoogLeNet [58]	91.70	91.66	89.36	**92.89**	71.70	86.84	56.58	21.95	75.33
VGGNet-16 [59]	88.37	97.38	85.55	89.88	78.90	95.46	72.38	**48.86**	82.10
ResNet50 [60]	86.88	92.29	89.07	92.07	72.55	96.51	59.07	21.60	76.25
Vit-base [61]	91.20	92.15	90.27	89.27	74.11	82.78	68.14	47.89	79.48
PVT-S [62]	92.41	**97.87**	84.33	91.42	76.42	96.36	69.72	24.28	79.10
MF^{2}CNet [64]	90.73	88.70	82.88	83.57	20.45	13.33	37.43	1.69	52.35
GRN-SNDL [65]	92.15	97.34	84.47	88.96	72.47	99.70	69.01	25.34	78.68
SE-Net [66]	81.56	89.30	77.36	88.62	52.30	89.84	46.49	20.47	68.24
SKAL [67]	89.95	89.42	77.63	88.68	56.74	81.35	55.35	17.98	69.64
LCNN-BFF [68]	90.74	90.87	87.96	90.53	60.17	**99.76**	55.77	25.59	75.17
SAGN [53]	90.91	96.22	83.70	90.53	**80.06**	**99.76**	69.90	26.32	79.67
PFFGCN [63]	91.18	92.20	**90.35**	92.32	70.93	99.05	69.65	24.62	78.79
FSCNet [69]	91.68	94.21	89.04	92.73	74.64	94.74	**74.69**	25.20	79.62
FAGN(Ours)	**94.06**	97.48	88.08	91.46	78.16	99.12	71.95	48.33	**83.58**

D. Ablation Study

As mentioned in Section IV, our FAGN model mainly consists of a feature extraction module (CNN) and a feature aggregation module (GCN). The feature aggregation module (GCN) constructs graph structures based on position correlation and semantic similarity. To study their influence on the FAGN model, we performed the following ablation experiments. First, we constructed four networks as follows.

1) Net-1: feature extraction module (CNN).

2) Net-2: feature extraction module (CNN) + feature aggregation module (spatial graph).

3) Net-3: feature extraction module (CNN) + feature aggregation module (semantic graph).

4) Net-4: feature extraction module (CNN) + feature aggregation module (spatial graph + semantic graph).

Here, the feature extraction module (CNN) is the DenseNet-201 network. Feature aggregation module (spatial graph) is the GCN that constructs the graph structure based only on position correlation. Feature aggregation module (semantic graph) is the GCN that constructs the graph structure based only on semantic similarity, and Net-4 is our FAGN model.

Then, we calculate their performance on the HRSD dataset to study the behavior of different models. The experimental setup is the same as mentioned in Section V-A. When the number ratio of training data to test data is 1:9, the accuracy of the four models on each class is shown in Fig. 7. From the observation, compared to the other three models, the performance of Net-4 has been improved in the "information loss," "cloud" and "band anomaly" categories. When the proportion of participating training data is 20%, the accuracy of the four models on each class is shown in Fig. 8. By observing Fig. 8, we can find that Net-4 performs well in each category, especially showing a significant improvement in the "other defects" category.

Since the performance growth in some categories is not obvious or even declines in Figs. 7 and 8, we also provide the OA values obtained by the four models, as shown in

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TABLE VII
OA OF THE FOUR NETS ON THE HRSD DATASET

Models	OA (1:9)	OA (2:8)
Net-1	85.44	88.02
Net-2	85.97	89.11
Net-3	81.84	83.23
Net-4	**86.54**	**90.13**

Table VII. As in Table IV, the best results in Table VII are shown in bold. OA(1:9) and OA(2:8) in this table are the same as in Table IV. From the observation, we can find that Net-4 achieves the best performance on both training sets. Compared with Net-1, when the training data ratio is 10%, Net-4 achieves an improvement of 1.10%, and when the training data ratio is 20%, Net-4 achieves an enhancement of 2.11%. These encouraging experimental results prove the effectiveness of the FAGN model. In addition, compared with Net-2, Net-4 achieved 0.57% and 1.02% improvement on the two training sets, respectively. This proves that graph construction based on semantic similarity is effective. Compared with Net-3, Net-4 achieved 4.70% and 6.90% enhancement on the two datasets, respectively. This demonstrates the importance of graph construction based on position correlation.

VI. CONCLUSION

In this article, we propose a dataset HRSD for remote sensing image defect detection, which is the first large-scale high-resolution remote sensing image defect detection dataset, covering various common defect types in remote sensing images. The HRSD dataset can help the remote sensing community further explore deep learning-based remote sensing image defect detection methods. Based on the HRSD dataset, to realize the detection of defects quickly, i.e., defect localization and classification, we directly perform defect classification on image patches. Furthermore, we propose a baseline model called FAGN, which fully utilizes the distribution characteristics of defects to achieve accurate defect detection. Extensive experiments on the HRSD dataset demonstrate the effectiveness of FAGN. Moreover, we will release the HRSD dataset and FAGN model to promote the development of remote sensing image defect detection.

REFERENCES

[1] Q. Zhao et al., "An overview of the applications of Earth observation satellite data: Impacts and future trends," *Remote Sens.*, vol. 14, no. 8, p. 1863, Apr. 2022.

[2] M. Chi, A. Plaza, J. A. Benediktsson, Z. Sun, J. Shen, and Y. Zhu, "Big data for remote sensing: Challenges and opportunities," *Proc. IEEE*, vol. 104, no. 11, pp. 2207–2219, Nov. 2016.

[3] Y. Ma et al., "Remote sensing big data computing: Challenges and opportunities," *Future Gener. Comput. Syst.*, vol. 51, pp. 47–60, Oct. 2015.

[4] D. Radočaj, M. Jurićić, and M. Gaćparović, "The role of remote sensing data and methods in a modern approach to fertilization in precision agriculture," *Remote Sens.*, vol. 14, no. 3, p. 778, Feb. 2022.

[5] A. Teodoro and L. Duarte, "The role of satellite remote sensing in natural disaster management," in *Nanotechnology-Based Smart Remote Sensing Networks for Disaster Prevention*. Amsterdam, The Netherlands: Elsevier, 2022, pp. 189–216.

[6] P. Zhong and R. Wang, "A multiple conditional random fields ensemble model for urban area detection in remote sensing optical images," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 12, pp. 3978–3988, Dec. 2007.

[7] L. Durieux, E. Lagabrielle, and A. Nelson, "A method for monitoring building construction in urban sprawl areas using object-based analysis of spot 5 images and existing GIS data," *ISPRS J. Photogramm. Remote Sens.*, vol. 63, no. 4, pp. 399–408, Jul. 2008.

[8] Y. Bai, S. Shao, S. Zhao, W. Liu, D. Tao, and B. Liu, "EME: Energy-based multiexpert model for long-tailed remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4701812.

[9] J. Wu, C. Lang, G. Cheng, X. Xie, and J. Han, "Retentive compensation and personality filtering for few-shot remote sensing object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 7, pp. 5805–5817, Jul. 2024.

[10] Y. Pang, L. Yao, Y. Luo, C. Dong, Q. Kong, and B. Chen, "RepSViT: An efficient vision transformer based on spiking neural networks for object recognition in satellite on-orbit remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5610916.

[11] Y. Chang, L. Yan, H. Fang, and C. Luo, "Anisotropic spectral–spatial total variation model for multispectral remote sensing image destriping," *IEEE Trans. Image Process.*, vol. 24, no. 6, pp. 1852–1866, Jun. 2015.

[12] Y.-J. Sun, T.-Z. Huang, T.-H. Ma, and Y. Chen, "Remote sensing image stripe detecting and destriping using the joint sparsity constraint with iterative support detection," *Remote Sens.*, vol. 11, no. 6, p. 608, Mar. 2019.

[13] P. Li, L. Dong, H. Xiao, and M. Xu, "A cloud image detection method based on SVM vector machine," *Neurocomputing*, vol. 169, pp. 34–42, Dec. 2015.

[14] A. Fisher, "Cloud and cloud-shadow detection in SPOT5 HRG imagery with automated morphological feature extraction," *Remote Sens.*, vol. 6, no. 1, pp. 776–800, Jan. 2014.

[15] S. Le Hégarat-Mascle and C. André, "Use of Markov random fields for automatic cloud/shadow detection on high resolution optical images," *ISPRS J. Photogramm. Remote Sens.*, vol. 64, no. 4, pp. 351–366, Jul. 2009.

[16] F. Tsai and W. W. Chen, "Striping noise detection and correction of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 12, pp. 4122–4131, Dec. 2008.

[17] E. Baseski and Ç. Cenaras, "Texture and color based cloud detection," in *Proc. 7th Int. Conf. Recent Adv. Space Technol. (RAST)*, Jun. 2015, pp. 311–315.

[18] J. H. Jeppesen, R. H. Jacobsen, F. Inceoglu, and T. S. Toftegaard, "A cloud detection algorithm for satellite imagery based on deep learning," *Remote Sens. Environ.*, vol. 229, pp. 247–259, May 2019.

[19] J. Yu, Y. Li, X. Zheng, Y. Zhong, and P. He, "An effective cloud detection method for Gaofen-5 images via deep learning," *Remote Sens.*, vol. 12, no. 13, p. 2106, Jul. 2020.

[20] K. Hu, D. Zhang, M. Xia, M. Qian, and B. Chen, "LCDNet: Light-weighted cloud detection network for high-resolution remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 4809–4823, 2022.

[21] C. Zhao, X. Zhang, N. Kuang, H. Luo, S. Zhong, and J. Fan, "Boundary-aware bilateral fusion network for cloud detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5403014.

[22] X. Li, X. Yang, X. Li, S. Lu, Y. Ye, and Y. Ban, "GCDB-UNet: A novel robust cloud detection approach for remote sensing images," *Knowl.-Based Syst.*, vol. 238, Feb. 2022, Art. no. 107890.

[23] B. Li et al., "Stripe noise detection of high-resolution remote sensing images using deep learning method," *Remote Sens.*, vol. 14, no. 4, p. 873, Feb. 2022.

[24] B. Li et al., "Synthesis and detection algorithms for oblique stripe noise of space-borne remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 3360268.

[25] J. Yang, J. Guo, H. Yue, Z. Liu, H. Hu, and K. Li, "CDnet: CNN-based cloud detection for remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 6195–6211, Aug. 2019.

[26] A. Bochkovskiy, C.-Y. Wang, and H.-Y. M. Liao, "YOLOv4: Optimal speed and accuracy of object detection," 2020, *arXiv:2004.10934*.

[27] Z. Li, H. Shen, H. Li, G. Xia, P. Gamba, and L. Zhang, "Multi-feature combined cloud and cloud shadow detection in GaoFen-1 wide field of view imagery," *Remote Sens. Environ.*, vol. 191, pp. 342–358, Mar. 2017.

[28] G. Zhang, X. Gao, Y. Yang, M. Wang, and S. Ran, "Controllably deep supervision and multi-scale feature fusion network for cloud and snow detection based on medium- and high-resolution imagery dataset," *Remote Sens.*, vol. 13, no. 23, p. 4805, Nov. 2021.

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[29] G. Zhang et al., "A multi-task driven and reconfigurable network for cloud detection in cloud-snow coexistence regions from very-high-resolution remote sensing images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 114, Nov. 2022, Art. no. 103070.

[30] X. Wu, Z. Shi, and Z. Zou, "A geographic information-driven method and a new large scale dataset for remote sensing cloud/snow detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 174, pp. 87–104, Apr. 2021.

[31] Q. He, X. Sun, Z. Yan, and K. Fu, "DABNet: Deformable contextual and boundary-weighted network for cloud detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2021.

[32] Z. Shao, J. Deng, L. Wang, Y. Fan, N. Sumari, and Q. Cheng, "Fuzzy AutoEncode based cloud detection for remote sensing imagery," *Remote Sens.*, vol. 9, no. 4, p. 311, Mar. 2017.

[33] Z. Yan et al., "Cloud and cloud shadow detection using multilevel feature fused segmentation network," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 10, pp. 1600–1604, Oct. 2018.

[34] Z. Shao, Y. Pan, C. Diao, and J. Cai, "Cloud detection in remote sensing images based on multiscale features-convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 6, pp. 4062–4076, Jun. 2019.

[35] M. Luotamo, S. Metsämäki, and A. Klami, "Multiscale cloud detection in remote sensing images using a dual convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 4972–4983, Jun. 2021.

[36] Z. Zhang, S. Yang, S. Liu, B. Xiao, and X. Cao, "Ground-based cloud detection using multiscale attention convolutional neural network," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2021.

[37] N. Ma, L. Sun, Y. He, C. Zhou, and C. Dong, "CNN-TransNet: A hybrid CNN-transformer network with differential feature enhancement for cloud detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[38] R. Couturier, H. N. Noura, O. Salman, and A. Sider, "A deep learning object detection method for an efficient clusters initialization," 2021, *arXiv:2104.13634*.

[39] A. M. Ikotun, A. E. Ezugwu, L. Abualigah, B. Abuhaija, and J. Heming, "K-means clustering algorithms: A comprehensive review, variants analysis, and advances in the era of big data," *Inf. Sci.*, vol. 622, pp. 178–210, Apr. 2023.

[40] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," 2016, *arXiv:1609.02907*.

[41] R. Ying, R. He, K. Chen, P. Eksombatchai, W. L. Hamilton, and J. Leskovec, "Graph convolutional neural networks for web-scale recommender systems," in *Proc. 24th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, 2018, pp. 974–983.

[42] R. Dai, S. Xu, Q. Gu, C. Ji, and K. Liu, "Hybrid spatio-temporal graph convolutional network: Improving traffic prediction with navigation data," in *Proc. 26th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Aug. 2020, pp. 3074–3082.

[43] J. Qiu, J. Tang, H. Ma, Y. Dong, K. Wang, and J. Tang, "DeepInf: Social influence prediction with deep learning," in *Proc. 24th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, 2018, pp. 2110–2119.

[44] L. Yao, C. Mao, and Y. Luo, "Graph convolutional networks for text classification," in *Proc. AAAI Conf. Artif. Intell.*, 2019, vol. 33, no. 1, pp. 7370–7377.

[45] J. Chen and X. Wang, "Open set few-shot remote sensing scene classification based on a multiorder graph convolutional network and domain adaptation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4709517.

[46] Y. Gao, J. Shi, J. Li, and R. Y. Wang, "Remote sensing scene classification based on high-order graph convolutional network," *Eur. J. Remote Sens.*, vol. 54, no. 1, pp. 141–155, 2021.

[47] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2020.

[48] L. Mou, X. Lu, X. Li, and X. X. Zhu, "Nonlocal graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8246–8257, Dec. 2020.

[49] S. Wan, C. Gong, P. Zhong, S. Pan, G. Li, and J. Yang, "Hyperspectral image classification with context-aware dynamic graph convolutional network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 597–612, Jan. 2021.

[50] X. He, Y. Chen, and P. Ghamisi, "Dual graph convolutional network for hyperspectral image classification with limited training samples," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2021.

[51] K. Xu, H. Huang, P. Deng, and Y. Li, "Deep feature aggregation framework driven by graph convolutional network for scene classification in remote sensing," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 10, pp. 5751–5765, Oct. 2022.

[52] X. Zhang, X. Tan, G. Chen, K. Zhu, P. Liao, and T. Wang, "Object-based classification framework of remote sensing images with graph convolutional networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2021.

[53] Y. Yang, X. Tang, Y.-M. Cheung, X. Zhang, and L. Jiao, "SAGN: Semantic-aware graph network for remote sensing scene classification," *IEEE Trans. Image Process.*, vol. 32, pp. 1011–1025, 2023.

[54] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 8024–8035.

[55] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 4700–4708.

[56] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 248–255.

[57] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.

[58] C. Szegedy et al., "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2015, pp. 1–9.

[59] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.

[60] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[61] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[62] W. Wang et al., "Pyramid vision transformer: A versatile backbone for dense prediction without convolutions," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 568–578.

[63] C. Zhang and B. Wang, "Progressive feature fusion framework based on graph convolutional network for remote sensing scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 3270–3284, 2024.

[64] L. Bai, Q. Liu, C. Li, Z. Ye, M. Hui, and X. Jia, "Remote sensing image scene classification using multiscale feature fusion covariance network with octave convolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5620214.

[65] J. Kang, R. Fernandez-Beltran, D. Hong, J. Chanussot, and A. Plaza, "Graph relation network: Modeling relations between scenes for multilabel remote-sensing image classification and retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4355–4369, May 2021.

[66] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2018, pp. 7132–7141.

[67] Q. Wang, W. Huang, Z. Xiong, and X. Li, "Looking closer at the scene: Multiscale representation learning for remote sensing image scene classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 4, pp. 1414–1428, Apr. 2022.

[68] C. Shi, T. Wang, and L. Wang, "Branch feature fusion convolution network for remote sensing scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 5194–5210, 2020.

[69] W. Wang, Y. Sun, J. Li, and X. Wang, "Frequency and spatial based multi-layer context network (FSCNet) for remote sensing scene classification," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 128, Apr. 2024, Art. no. 103781.

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